

Knapsack Problem

$n=7$
 $m=15$

Objects:	0	1	2	3	4	5	6	7	
profits: P		10	5	15	7	6	18	3	<u>Constraint</u>
weights: w		2	3	5	7	1	4	1	$\sum x_i w_i \leq m$
$\frac{P}{w}$		5	1.3	3	1	6	4.5	3	<u>Objective</u>
x		$(1$	$\frac{2}{3}$	1	0	1	1	$1)$	$\boxed{\max \sum x_i p_i}$
		x_1	x_2	x_3	x_4	x_5	x_6	x_7	

$$\sum x_i w_i = 1 \times 2 + \frac{2}{3} \times 3 + 1 \times 5 + 0 \times 7 + 1 \times 1 + 1 \times 4 + 1 \times 1$$

$$2 + 2 + 5 + 0 + 1 + 4 + 1 = 15$$

$$\sum x_i p_i = 1 \times 10 + \frac{2}{3} \times 5 + 1 \times 15 + 1 \times 6 + 1 \times 18 + 1 \times 3$$

$$= 10 + 2 \times 1.3 + 15 + 6 + 18 + 3 = 54.6$$